Chapter 10. Estimating Population Size and Nonresponse

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1 Capture–Recapture Estimation of N

To estimate the number N of fish in a lake, mark and release n_1 fish, then catch n_2 and count the m marked ones.

Definition 10.1: Assumptions

1. The population is closed (no births, deaths or migration between the samples).
2. Each sample is an SRS from the population (marking does not change catchability).
3. The two samples are independent.

The proportion of marked fish in the second sample should resemble that in the population:

$$\frac{m}{n_2} \approx \frac{n_1}{N} \quad \Longrightarrow \quad \hat{N} = \frac{n_1 n_2}{m}$$

(the Lincoln–Petersen estimator).

1.1 Capture–recapture as a ratio estimator

Theorem 10.2

$\hat{N}$ is a ratio estimator of a population total, and

$$\widehat{\mathrm{Var}}(\hat{N}) = \frac{n_1^2 n_2^2 (n_2 - m)}{m^3 (n_2 - 1)} \approx \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

Define for every element $y_i = 1$, so $\tau_y = \sum_{i=1}^{N} y_i = N$, and $x_i = 1$ if element i is marked, 0 otherwise, so $\tau_x = \sum_{i=1}^{N} x_i = n_1$ is known. The recapture sample of size n_2 gives

$$r = \frac{\sum_{i=1}^{n_2} y_i}{\sum_{i=1}^{n_2} x_i} = \frac{n_2}{m}, \qquad \hat{N} = \hat{\tau}_y = r\tau_x = \frac{n_2}{m} n_1.$$

From Ch. 5, $\widehat{\mathrm{Var}}(\hat{N}) = \tau_x^2 \widehat{\mathrm{Var}}(r) = n_1^2 \widehat{\mathrm{Var}}(r)$ with

$$\widehat{\mathrm{Var}}(r) = \frac{1}{\bar{x}^2} \cdot \frac{N - n_2}{N} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \sum_{i=1}^{n_2} (y_i - r x_i)^2,$$

and we take the fpc ≈ 1. Now

$$\bar{x} = \frac{m}{n_2}, \qquad \sum_{i=1}^{n_2} (y_i - r x_i)^2 = \sum y_i^2 - 2r \sum x_i y_i + r^2 \sum x_i^2 = n_2 - 2\frac{n_2}{m} m + \frac{n_2^2}{m^2} m = \frac{n_2(n_2 - m)}{m},$$

using $y_i^2 = y_i = 1$ and $x_i^2 = x_i$. Therefore

$$\widehat{\mathrm{Var}}(r) = \frac{n_2^2}{m^2} \cdot \frac{1}{n_2} \cdot \frac{1}{n_2 - 1} \cdot \frac{n_2(n_2 - m)}{m} = \frac{n_2^2 (n_2 - m)}{m^3 (n_2 - 1)},$$

and multiplying by n_1^2 gives the result.

1.2 Maximum likelihood

Definition 10.3

For the joint pdf $f(\mathbf{x} \mid \theta)$, $L(\theta \mid \mathbf{x}) = f(\mathbf{x} \mid \theta)$ is the **likelihood function**; $L(\theta_1 \mid \mathbf{x}) \geq L(\theta_2 \mid \mathbf{x})$ makes θ_1 more plausible. The **maximum likelihood estimator** is $\hat{\theta}(\mathbf{x}) = \arg\max_{\theta \in \Theta} L(\theta \mid \mathbf{x})$.

Example 10.4: Normal mean

For $X_1, \ldots, X_n \overset{\text{iid}}{\sim} N(\theta, 1)$,

$$\ell(\theta) = \log L(\theta \mid \mathbf{x}) = -\frac{n}{2} \log 2\pi - \frac{1}{2} \sum_{i=1}^{n} (x_i - \theta)^2, \qquad \ell'(\hat{\theta}) = \sum (x_i - \hat{\theta}) = 0 \;\Rightarrow\; \hat{\theta} = \bar{x}.$$

Theorem 10.5: Large-sample properties of the MLE

Under regularity conditions,

1. (consistency) $\lim_{n \to \infty} P(|\hat{\theta}_n - \theta| \geq \varepsilon) = 0$ for every $\varepsilon > 0$;
2. (asymptotic normality) $\sqrt{n}(\hat{\theta}_n - \theta) \to N(0, 1/I(\theta))$, i.e. $\hat{\theta}_n \approx N(\theta, \frac{1}{nI(\theta)})$, where $I(\theta) = \mathrm{E}\big[\big(\frac{\partial}{\partial \theta} \log f(X \mid \theta)\big)^2\big]$ is the Fisher information of one observation.

Theorem 10.6: MLE of N

Model the recapture sample as $X_i = 1$ (marked) or 0, $X_i \overset{\text{iid}}{\sim} \mathrm{Ber}(p)$ with $p = n_1/N$, $i = 1, \ldots, n_2$. Then

$$\hat{N}_{\mathrm{MLE}} = \frac{n_1 n_2}{m}, \qquad \mathrm{Var}(\hat{N}_{\mathrm{MLE}}) \approx \frac{N^2 (N - n_1)}{n_1 n_2}, \qquad \widehat{\mathrm{Var}}(\hat{N}_{\mathrm{MLE}}) = \frac{n_1^2 n_2 (n_2 - m)}{m^3}.$$

Derivation

With $\sum_{i=1}^{n_2} X_i = m$,

$$L(N) = \prod_{i=1}^{n_2} \Big(\frac{n_1}{N}\Big)^{X_i}\Big(1-\frac{n_1}{N}\Big)^{1-X_i} = \Big(\frac{n_1}{N}\Big)^{m}\Big(1-\frac{n_1}{N}\Big)^{n_2-m},$$

$$\ell(N) = m\log n_1 - m\log N + (n_2-m)\log(N-n_1) - (n_2-m)\log N, \qquad \ell'(N) = \frac{n_2-m}{N-n_1} - \frac{n_2}{N}.$$

Setting $\ell'(N)=0$: $N(n_2-m) = n_2(N-n_1)$, so $\hat N = n_1 n_2/m$.

Fisher information. For one observation,

$$\frac{\partial}{\partial N}\log f(x\mid N) = \frac{1-x}{N-n_1} - \frac{1}{N} = \frac{n_1/N - x}{N-n_1} = \frac{p-x}{N-n_1},$$

$$I(N) = \mathrm{E}\Big(\frac{p-X}{N-n_1}\Big)^2 = \frac{p(1-p)}{(N-n_1)^2} = \frac{n_1}{N^2(N-n_1)}.$$

Hence $\mathrm{Var}(\hat N_{\mathrm{MLE}}) \approx [n_2 I(N)]^{-1} = \frac{N^2(N-n_1)}{n_1 n_2}$. Plugging in $\hat N = n_1 n_2/m$:

$$\frac{(n_1n_2/m)^2\,(n_1n_2/m - n_1)}{n_1n_2} = \frac{n_1^2n_2^2}{m^2}\cdot\frac{n_1(n_2-m)}{m}\cdot\frac{1}{n_1n_2} = \frac{n_1^2n_2(n_2-m)}{m^3}.$$

The ratio-estimator and likelihood approaches agree: $\widehat{\mathrm{Var}}(\hat\tau_y) \approx \widehat{\mathrm{Var}}(\hat N_{\mathrm{MLE}})$.

Example 10.7: Illustration

If $n_1 = 200$ fish are marked and a second catch of $n_2 = 100$ contains $m = 20$ marked fish, then $\hat N = 200\cdot 100/20 = 1000$ and

$$\widehat{\mathrm{Var}}(\hat N) = \frac{200^2\cdot 100\cdot 80}{20^3} = 40{,}000, \qquad 2\sqrt{40000} = 400,$$

so N is estimated as 1000 ± 400.

2 Nonresponse

Nonresponse (missing data) biases the estimation of population parameters. Divide the population into a response stratum R and a nonresponse stratum:

	size	mean
Respondents	N_1	μ_1
Nonrespondents	N_2	μ_2
	$N = N_1 + N_2$	μ

Theorem 10.8: Nonresponse bias

If $\bar y_1$ is unbiased for the respondents' mean μ_1, then as an estimator of μ,

$$\mathrm{E}(\bar y_1) - \mu = \frac{N_2}{N}(\mu_1 - \mu_2).$$

Proof. $\mu = \frac{N_1}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \frac{N_1+N_2}{N}\mu_1 - \frac{N_2}{N}\mu_1 + \frac{N_2}{N}\mu_2 = \mu_1 - \frac{N_2}{N}(\mu_1-\mu_2)$. ∎

The bias is small only if N_2/N is small or $\mu_1 \approx \mu_2$; larger samples do not reduce it.

2.1 A conservative interval for a proportion

Suppose n_1 respondents are an SRS from the N_1 respondents of the population, so a 95% interval for their proportion p_1 is

$$\hat{p}_1 \pm 2\sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1-1}\cdot\frac{N_1-n_1}{N_1}}.$$

Only $0 \le p_2 \le 1$ is known. With $w_h = N_h/N$, $p = w_1p_1 + w_2p_2$; take $p_2 = 0$ and $p_2 = 1$:

$$\hat{p}_L = w_1\Big(\hat{p}_1 - 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\Big) + w_2\cdot 0,$$
$$\hat{p}_U = w_1\Big(\hat{p}_1 + 2\sqrt{\tfrac{\hat{p}_1\hat{q}_1}{n_1-1}\tfrac{N_1-n_1}{N_1}}\Big) + w_2\cdot 1.$$

If the interval covers p_1, $[\hat{p}_L, \hat{p}_U]$ covers p for any p_2, so $P(\hat{p}_L \le p \le \hat{p}_U) \ge 0.95$. The width grows by w_2: reducing nonresponse matters more than increasing n.

3 Selected exercises

Motivation. Lincoln–Petersen as completion of a 2×2 table and from the exact hypergeometric likelihood.

Exercise 10.1 (Deer population)

A random sample of $n_1 = 300$ deer is captured, tagged and released. A second sample of $n_2 = 200$ taken two weeks later contains $m = 62$ tagged deer.
(a) Arrange the data as a 2×2 table (caught or not in each sample) and estimate the missing cell, assuming independent captures.
(b) Show that the resulting estimate of N is the Lincoln–Petersen estimator, and place a bound on the error.

Solution.

(a) The table has one unobservable cell x (deer caught in neither sample):

	in 2nd sample	not in 2nd	total
in 1st sample	62	238	300
not in 1st	138	x	
total	200		N

Independence of the two captures means that the odds ratio of the table is 1, i.e. $62\,x = 238\times 138$, so $\hat{x} = 32{,}844/62 = 529.7$.

(b)

Step 1. In general, with cells $n_{11} = m$, $n_{12} = n_1 - m$, $n_{21} = n_2 - m$ and $\hat{x} = n_{12}n_{21}/n_{11}$,

$$\hat{N} = n_{11} + n_{12} + n_{21} + \frac{n_{12}n_{21}}{n_{11}} = \frac{(n_{11}+n_{12})(n_{11}+n_{21})}{n_{11}} = \frac{n_1 n_2}{m}.$$

Step 2. Here $\hat{N} = 62 + 238 + 138 + 529.7 = 967.7 = 300\times 200/62$.

Step 3. By the MLE theorem, $\widehat{\mathrm{Var}}(\hat N) = \dfrac{n_1^2 n_2 (n_2 - m)}{m^3} = \dfrac{300^2 \cdot 200 \cdot 138}{62^3} = 10{,}423$, and $2\sqrt{10{,}423} = 204$. The herd is estimated at 968 ± 204 deer. ■

Exercise 10.2 (The hypergeometric likelihood)

If the second sample is an SRS of n_2 drawn without replacement from N animals of which n_1 are marked, then

$$L(N) = P(m \mid N) = \frac{\binom{n_1}{m}\binom{N-n_1}{n_2-m}}{\binom{N}{n_2}}, \qquad N \geq n_1 + n_2 - m.$$

Assuming $m \geq 1$, show that the MLE of N over the integers is $\lfloor n_1 n_2 / m \rfloor$.

Solution. [Supplement]

Step 1. (Likelihood ratio.) For $N - 1 \geq n_1 + n_2 - m$, using $\binom{a}{c}/\binom{a-1}{c} = \frac{a}{a-c}$,

$$\frac{L(N)}{L(N-1)} = \frac{\binom{N-n_1}{n_2-m}}{\binom{N-1-n_1}{n_2-m}} \cdot \frac{\binom{N-1}{n_2}}{\binom{N}{n_2}} = \frac{N-n_1}{N-n_1-n_2+m} \cdot \frac{N-n_2}{N}.$$

Step 2. (Sign.) Both denominators are positive, so $L(N) \geq L(N-1)$ iff $(N-n_1)(N-n_2) \geq N(N-n_1-n_2+m)$. Expanding, the terms N^2 and $-N(n_1+n_2)$ cancel and the condition becomes $n_1 n_2 \geq N m$, i.e. $N \leq n_1 n_2 / m$.

Step 3. (Maximum.) L is nondecreasing for $N \leq n_1 n_2 / m$ and strictly decreasing after, so maximal at $\lfloor n_1 n_2 / m \rfloor$ (tied with $n_1 n_2 / m - 1$ if integer), which is $\geq n_1 + n_2 - m$ since $(n_1 - m)(n_2 - m) \geq 0$: the Lincoln–Petersen estimate. ■